



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Electronics and Telecommunication Engineering

Summer Term Examination

Academic Year: 2024-25

Class: TY B.Tech	Subject: Digital Signal Processing	Subject Code: PCC-EC318
Date: 19.06.2025	Time: 10:00 to 13:00	Max. Marks: 100

Note:

- This question paper contains 2 pages.*
- Assume suitable data whenever necessary and state it clearly.*
- Mobile phones and programmable calculators are strictly prohibited.*
- Exchange/Sharing of stationery, calculator etc. not allowed.*
- Figures to right indicate full marks.*
- The symbols carry usual meaning.*

Q.N.	Questions	Marks	CO	BT
Q.1	1) State the sampling theorem and determine the Nyquist rate and Nyquist interval of the following discrete time signal $X(n) = 4 + \cos(4000\pi n) + \sin(6000\pi n)$ 2) Find the linear convolution of the following sequence using graphical method (Flip, shift and multiply) $x(n) = \{4, 2, 1, 3\}; \quad h(n) = \{1, 2, 2, 1\}$	10	CO1	BT2, BT3
Q.2	1) For the sequence $x(n) = \begin{cases} 2^n & n < 0 \\ (1/3)^n & n = 0, 2, 4, \dots \\ (1/4)^n & n = 1, 3, 5, \dots \end{cases}$ Determine the z transform and plot the respective region of convergence. 2) Two casual discrete time signals $x[n]$ and $y[n]$ are related as $y[n] = \sum_{m=0}^n x[m]$. If the z Transform of $y[n]$ is $2 / z (z-1)^2$. Determine the value of $x[2]$.	10	CO3	BT1, BT2, BT3
Q.3	Find the DTFT of the following sequence, 1) $x(n) = (1/2)^{n-2} u(n-2)$ 2) $x(n) = (1/2)^n \sin(n\pi/4) u(n)$	10	CO3	BT1
Q.4	Prove the following properties of DTFT. 1) Differentiation property 2) Time shifting property	10	CO3	BT1
Q.5	Differentiate between circular convolution and linear convolution. Compute the circular convolution of the following sequences and compare it with linear convolution $x(n) = \{1, 2, 1, 2\}; \quad h(n) = \{4, 3, 2, 1\}$	10	CO1	BT1

Q.6	Write any five properties of DFT?	10	CO3	BT1
Q.7	a) Find the 4 point DFT of $x(n) = \{1, -1, 2, -2\}$ b) Find the IDFT of $X(k) = \{4, 2, 0, 4\}$	10	CO3	BT2, BT5
Q.8	Compute the IDFT of the sequence $X(k) = \{4, 1-j2.414, 0, 1-j0.414, 0, 1+j0.414, 0, 1+j2.414\}$ Using DIF FFT algorithm.	10	C04	BT5
Q.9	An 8 point sequence is given by $x(n) = \{2, 2, 2, 2, 1, 1, 1, 1\}$. Compute the 8 point DFT of $x(n)$ by a) Radix-2 DIT FFT Algorithm b) Radix-2 DIF FFT Algorithm	20	C04	BT5

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyse, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO1	Understand discrete time signals, classification of discrete time signals, signal processing operations.
CO2	Understand Z-transform, ROC of the rational Z-transform, Inverse Z-transform, Z transform properties.
CO3	Understand Discrete time Fourier transform, Discrete Fourier Transform, Discrete Fourier Transform properties, Computation of the DFT of real sequences
CO4	Apply FFT Algorithms like Decimation in frequency and Decimation in Time algorithms.
CO5	Design the FIR and IIR filters.



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded

SUMMER TERM Examination (Semester-II/VI) 2024-25

Third Year B. Tech.

Electronics and Telecommunication Engineering

Elective: Probability and Statistics - P&S

PEC - EC305

Date: 20/06/2025

Time: 10:00 to 13:00

Max. Marks : 100

Note: i. Assume suitable data if necessary. ii. Use of non-programmable calculators is permitted. iii. This paper contains one page only. iv. Solve any five questions.

Q.N	Question	Marks	CO	BT
Q.1	i. State and prove Bayes Theorem. Why is it important in statistics? Provide an example of its application. ii. Calculate the mean, median, mode for the following data set: 3, 5, 7, 9, 11, 13, 15.	20	1, 2	1, 2, 5, 6
Q.2	(a) Define probability and explain the concept of Total probability with an example? (b) Find the probability of getting sum 12, 6 and 3 when two dice are rolled.	20	1, 4	2, 3
Q.3	i. Prove that $\sigma^2 = E(X^2) - (E(X))^2$ ii. Define a hypothesis test. Differentiate between Type I and Type II errors.	20	2, 4	2, 5
Q.4	There are N=3 distinct boxes and M=3 distinct objects. Each object is to be placed in one of the boxes. (i) Total Number of Ways to Distribute Objects. (ii) Number of ways to distribute M objects into N boxes such that no box is empty.	20	5	5, 6
Q.5	Define Conditional probability distribution and Conditional probability density functions and state their properties?	20	1, 2	1
Q.6	i. In a standard normal distribution, what is the probability that a randomly selected value is between -1 and 1? ii. If two events A and B are independent then $P(A \cup B) = ?$ Explain why?	20	1, 2	1

***** !!!!!!! All the very Best !!!!!!! *****

Details below are for Information only: Bloom's Taxonomy: BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyze, BT5- Evaluate, BT6- Create



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Electronics and Telecommunication Engineering
Summer Term Examination
Academic Year: 2024-25

Class: TY B.Tech.

Date: 20.06.2025, Fri

Subject: Signals and Systems

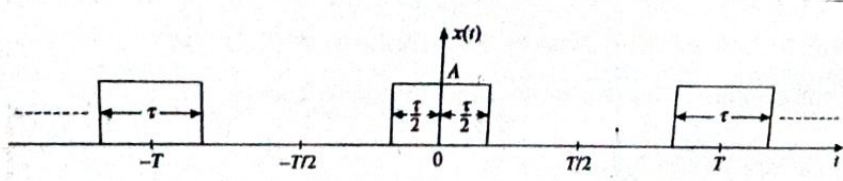
Time: 10:00 to 13:00

Subject Code: PCC-EC304

Max. Marks: 100

Note:

- Assume suitable data whenever necessary and state it clearly.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- All questions are compulsory.

Q.N.	Question	Marks	CO	BT
Q.1	Sketch the following signals. a) $r(t) u(t + 2)$ b) $r(-0.25t + 1)$	10	CO1	BT2
Q.2	Determine the power and rms value of the following signals. a) $x(t) = 12 \cos\left(20t + \frac{\pi}{3}\right) + 12 \sin\left(30t + \frac{\pi}{2}\right)$ b) $x(t) = e^{j2t} \cos 10t$	10	CO1	BT2
Q.3	Obtain the exponential Fourier series of the periodic gate function shown in the figure. Also plot amplitude and phase spectrum 	10	CO1	BT2
Q.4	Check whether the following system of equation is (with proper explanation) $y(t) = at^2 x(t) + bt x(t - 4)$ i) Static or Dynamic iii) Causal or Non-causal ii) Linear or Non-linear iv) Time-invariant or time variant	10	CO3	BT3
Q.5	Find the convolution of the following signals by graphical method $x(t) = e^{-3t}u(t)$ and $h(t) = u(t + 3)$	10	CO3	BT5
Q.6	Find the trigonometric Fourier series for the waveform shown in below figure.	15	CO2	BT4

Q.7	Find the Fourier transform of the following signal i) $e^{-t} \sin 5t u(t)$ ii) $x(t) = e^{- t }$ for $-2 < t < 2$ $= 0$ otherwise	10	CO2	BT5
Q.8	Find the Laplace transform of the following signals. a) $x(t) = e^{2t}u(t) + e^{-2t}u(-t)$ and find its ROC b) $x(t) = [1 + \sin 2t \cos 2t] u(t)$	15	CO2	BT4
Q.9	Prove the time scaling property and differentiation property of Fourier transform.	10	CO2	BT3

== ALL THE BEST ==

Bloom's Taxonomy (BT):

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes: On successful completion of this course, students should be able to:

CO 1	Analyze different types of signals.
CO 2	Represent continuous and discrete systems in time and frequency domain using different transforms.
CO 3	Investigate whether the system is stable.
CO 4	Sampling and reconstruction of a signal.